

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

A model estimates that at the end of each year from **2015** to **2020**, the number of squirrels in a population was **150%** more than the number of squirrels in the population at the end of the previous year. The model estimates that at the end of **2016**, there were **180** squirrels in the population. Which of the following equations represents this model, where n is the estimated number of squirrels in the population t years after the end of **2015** and $t \leq 5$?

Answer

- A. $n = 72(1.5)^t$
- B. $n = 72(2.5)^t$
- C. $n = 180(1.5)^t$
- D. $n = 180(2.5)^t$

Correct Answer: B

Rationale

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, **150%**, each year, the model can be represented by an exponential equation of the form $n = a\left(1 + \frac{p}{100}\right)^t$, where a is the estimated number of squirrels in the population at the end of **2015**, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was **150%** more than the number at the end of the previous year, $p = 150$. Substituting **150** for p in the equation $n = a\left(1 + \frac{p}{100}\right)^t$ yields $n = a\left(1 + \frac{150}{100}\right)^t$, which is equivalent to $n = a(1 + 1.5)^t$, or $n = a(2.5)^t$. It's given that the estimated number of squirrels at the end of **2016** was **180**. This means that when $t = 1$, $n = 180$. Substituting **1** for t and **180** for n in the equation $n = a(2.5)^t$ yields $180 = a(2.5)^1$, or $180 = 2.5a$. Dividing each side of this equation by **2.5** yields $72 = a$. Substituting **72** for a in the equation $n = a(2.5)^t$ yields $n = 72(2.5)^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was **150%** of, not **150%** more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of **2015**, not the end of **2016**, was **180**.